



Lesson 4 – What Is AI?

Algorithms and Predictions

Teacher-only slide – Lesson Overview

Lesson Preparation



- Print off IS IT AI? cards, one set each to a small group is appropriate – and keep the Teacher guide (with answers) for yourself (more details below).

IS IT AI? Student Sheet



Calculator	Mum's TikTok For You Page
Spotify personalised playlist	Smoke alarm
Face ID	Microwave timer
Google Maps directions	Amazon Alexa giving us a recipe
Traffic lights changing on timers	A robot trained to assemble tables
Tesla braking for pedestrian	A vending machine giving you change
Google showing search results	Siri suggesting a restaurant

IS IT AI? Teacher Guide



AI	NOT AI	DEBATABLE
Calculator NOT AI. Follows fixed rules — it does exactly what you type, every time. No learning, no patterns.	Mum's TikTok For You Page AI. Learns from what you watch, like, and skip to predict what to show next.	
Spotify personalised playlist AI. Analyzes your listening history and patterns from millions of users to predict songs you'll enjoy.	Smoke alarm NOT AI. A sensor detects smoke above a fixed threshold and triggers an alarm — a simple perceptron rule.	
Face ID AI. Uses a neural network trained on thousands of faces to recognise yours, even in different lighting.	Microwave timer NOT AI. Counts down a set time and stops — a fixed rule with no learning involved.	
Google Maps directions AI. Learns from real-time traffic data and millions of past trips to predict and reroute the fastest path.	Amazon Alexa giving us a recipe AI. Uses natural language processing to understand your question and retrieve a relevant answer.	
Traffic lights changing on timers NOT AI. Switches on a fixed schedule — it doesn't observe or adapt to traffic conditions.	A robot trained to assemble tables DEBATABLE. The word "trained" suggests AI, but most factory robots follow perceptron scripts. Ask: does it learn, or just repeat?	
Tesla braking for pedestrian DEBATABLE. Basic braking uses a sensor and a fixed rule (not AI). Full self-driving systems that identify what the obstacle is use AI.	A vending machine giving you change NOT AI. Calculates change using fixed math — no pattern recognition or learning involved.	
Google showing search results DEBATABLE. Basic keyword matching is not AI, but Google uses AI to rank results based on what people click on.	Siri suggesting a restaurant AI. Uses natural language understanding and learns from your habits and location to make personalised suggestions.	

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Lesson 4 - Overview



1. **Lesson 3 Recap** - 5 min
 2. **What is AI** – 10 min
 - 2.1 Pattern recognition
 - 2.2 A tool not a mind
 3. **Is it AI** – 25 min
 - 3.1 Students sorting cards
 - 3.2 Class discussion + clarification
 4. **Building Blossom's spine** - 30 min (+ any catch up 10 min)
 5. **Survey** - 5 min
 6. **Summary** - 5 min
- Total: 90 min

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Lesson 4 – Learning objectives



After this lesson, students will be able to:

- Define AI in their own words
- Identify at least three examples of AI they encounter in everyday life
- Explain the difference between a programmed rule and AI-driven decision making
- Describe AI as *pattern recognition* rather than human-like thinking or understanding
- Distinguish between AI and non-AI technology in a real-world scenario
- Reflect critically on the role AI already plays in their daily life

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Lesson 4 - Student success criteria



- I can explain what AI is in my own words
- I can give at least three examples of AI I use or see every day
- I can explain the difference between a programmed rule and AI
- I can describe AI as finding patterns rather than actually thinking or understanding
- I can look at a piece of technology and decide whether it uses AI or not
- I can reflect on how much AI is already part of my daily life

Teacher-only slide – Instruction Guide

Recap Coding

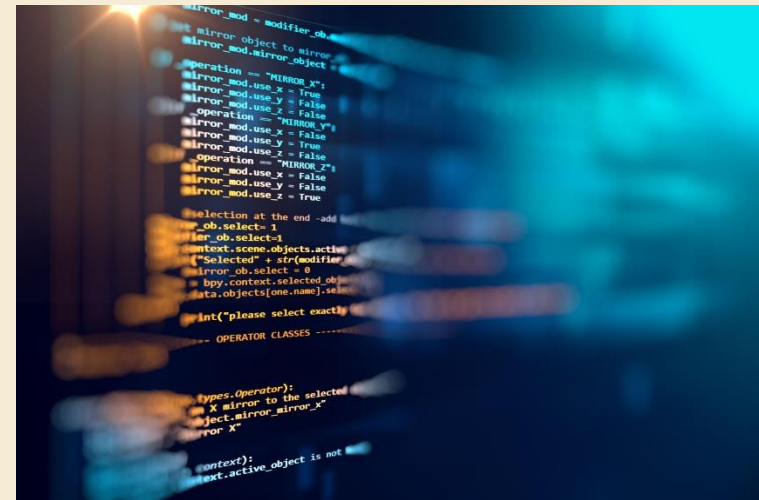


- **Instruction note: Cover the following concepts**
- *‘Code is just a language that robots and computers understand!’*
- *‘We use IF STATEMENTS to tell the computer to do something IF an event occurs’*
- *‘We use loops to tell our computer to continue to repeat a stack of code’*

Let's remember what we learned in Lesson 3!



- Give an example of a coding language
- How does an IF STATEMENT help us automate our code?
- How do loops help make our code shorter (or neater)



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Introduction to AI



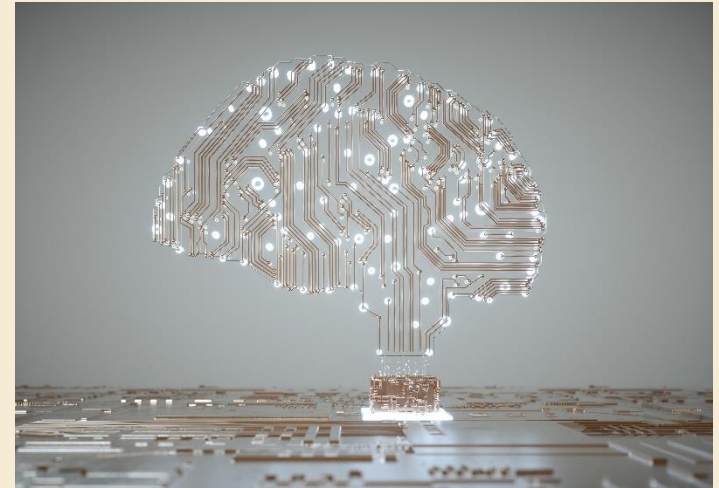
- **Sample script:** *“AI stands for artificial intelligence. It’s when computers are programmed to do things that would normally require thinking (like humans do). It learns from patterns in data, just like how we learn from our experiences. We know AI is everywhere already! It helps us with YouTube/Netflix suggestions, Siri in our phones, Google Translate, Maps and autocorrect.*
- *It is made by humans but doesn’t have emotions or feelings like we do. There are many different types of AI and we will learn about them in the next few weeks. You may have heard of:*
 - *generative AI: a tool that is able to generate text, images or videos*
 - *LLM (large language models) an AI tool that learns from and produces text, based on predictions*
 - *machine learning, enables systems to learn from many types of data! ”*

Note: Some products like ChatGPT use all of these AI technologies together.

Introduction to AI



- **AI = Artificial Intelligence**
- 🧠 Learns from patterns in data
- ⚙️ A tool made by humans — not magic!
- 🧩 Generative AI - LLMs - Machine Learning



Learning from patterns



- **Sample script:** *“If we went to our YouTube or Netflix recommendations we would see a whole bunch of TV shows, movies and videos to watch. How does the AI suggest we watch these? It doesn’t know us, and it’s definitely never watched TV with us, we might not have ever told it what we do and don’t like. It’s because it can look at data of millions of other people who have watched similar things to us and see what they watched next, it then suggests them as something we might like to watch next!*
- *In other words, it is using data to make a prediction”*

Learning from patterns—not thinking



- **AI finds patterns — it does NOT think like us**
- 📺 Netflix doesn't know you
- It spots what millions of similar people watched next
- ***Pattern → Prediction (not understanding!)***

It's EVERYWHERE!



Autocorrect

fixes typos as you
type



Face ID

recognises your
face



Spam filter

blocks junk mail



ChatGPT

predicts next word



Can you think of more examples?

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Is It AI?



- **Instruction note:** Some of the cards are debatable. Here are some discussion points to raise with the students:
- **Teslas** – They have precoded rules, if the ultrasonic sensor detects something in front of it at X distance it will brake (**this is not AI**). However, some newer models predict where the cars or objects in front will go next and brake accordingly (**this is AI**)
- **Google search** – Google will generate a list of results based on common words which is precoded and doesn't use AI, however newer models may predict what a person is looking for, based on what links others clicked on and show them first
- **Assembly robot** – Can use prediction models to recognise which pieces are likely to go where, however most are precoded and are not AI

IS IT AI?

Teacher Guide

AI

NOT AI

DEBATABLE

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Is it AI?



- We are going to do an activity, separating things into whether we think they are AI or not
- Remember, AI has access to billions of resources and makes predictions!
- Some of the cards may be debatable!
 - If you think the card is debatable, create an argument for why you think it is or isn't AI

Teacher-only slide – Instruction Guide



AI Mistakes and Hallucinations

- **Sample script:** *"Imagine you've read every bike review ever written — every magazine, website, and forum post — and you remember all of it. I ask you: 'What type of bike is a Talon?' Easy — it's a Giant mountain bike. You've seen that thousands of times. Confident, correct, done. Now I ask: 'What's better for a 12 year old — a Talon Pro 5000 or a Trek Thunderbolt X?' You've never seen those exact names. But you've read so much about similar bikes that you piece something together, you even produce a comparison table. Talon Pro 5000 comes out on top.*
- *Sounds great. Except — I made those bikes up. They don't exist. You filled in the gap with something that sounded completely plausible and presented it with total confidence.*
- *That's exactly what AI can do. We call it a hallucination — not because the AI is dreaming, but because it produces something that sounds real but isn't. And the scary part? It doesn't know it's wrong. No alarm goes off. It just gives you the most pattern-consistent answer it can. **So whenever AI gives you a confident, detailed answer, ask yourself: is this actually true, or does it just sound true?"***

Can it make mistakes?



-  **You've read every bike review ever written...**
- Easy: "What is a Talon?" → Giant mountain bike 
- Hard: "Talon Pro 5000 vs Trek Thunderbolt X?"
 - *Neither bike exists — but AI gave a confident answer!*
-  **This is a HALLUCINATION — always check AI answers!**

Questionnaire



Summary



- AI – Artificial Intelligence
- Come up with 3 different examples of AI in our everyday life
- Explain the difference between predicting and understanding
- Why is it always important to check the answers AI gives us?